Lecture 1: Introduction

41-501
Applied AI

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Class Organization

- Canvas as Learning Platform
  - Lecture files, lab code, lecture recordings
  - Announcements
- MS Teams for recordings and group chat
  - Lectures are being recorded
  - Labs are NOT being recorded (-> come to class)
- Assignments:
  - One assignment after the 4th lecture (19 September)
  - One assignment/presentation at the end (10 October)
- (No lecture on 26 September)

Overview

- Introduction
- Terminology
- History of AI
- Future of AI
- Lab: BYOC

Introduction

Introduce yourself:

- Your name
- How much do you understand AI?
- One tech tool or app you use regularly (except Chatbots)

About Me

- French & German
- Mechanical Engineering @ TUHH
- Formula Student Electric
- Master Thesis: Autonomous Driving
- PhD in Brain-inspired AI @ TUM
- Postdoc in Comp. Neuroscience @ UniMelb
- Now: Professor AI & Data Science @ CMKL

➢ Algorithmic Optimization and Energy Efficiency

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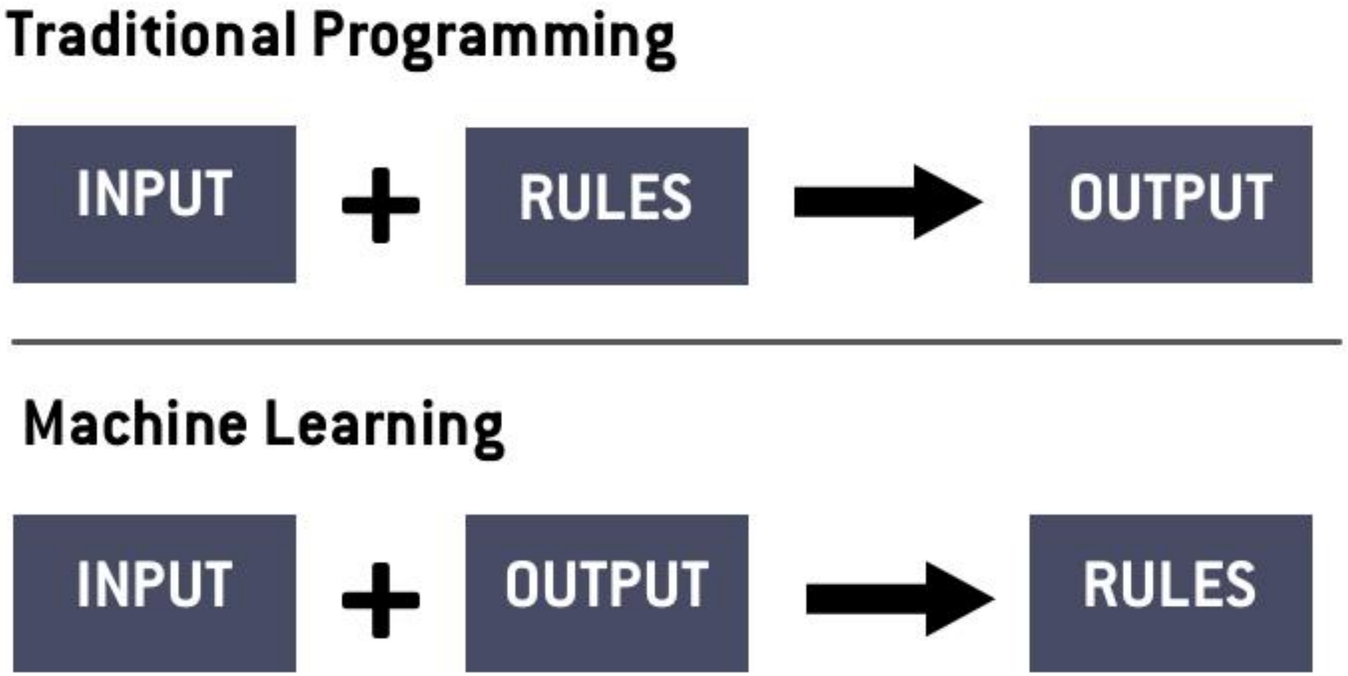
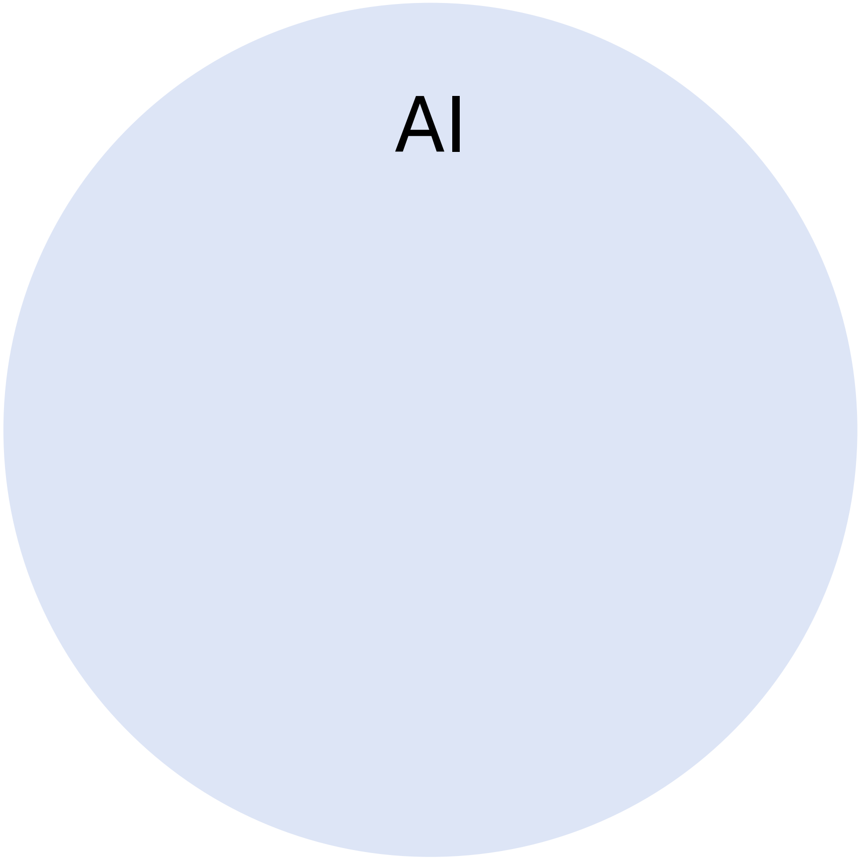
Traditional Programming vs ML



Terminology: Artificial Intelligence

Systems that can perform tasks requiring human-like intelligence, such as reasoning, perception, or decision-making.

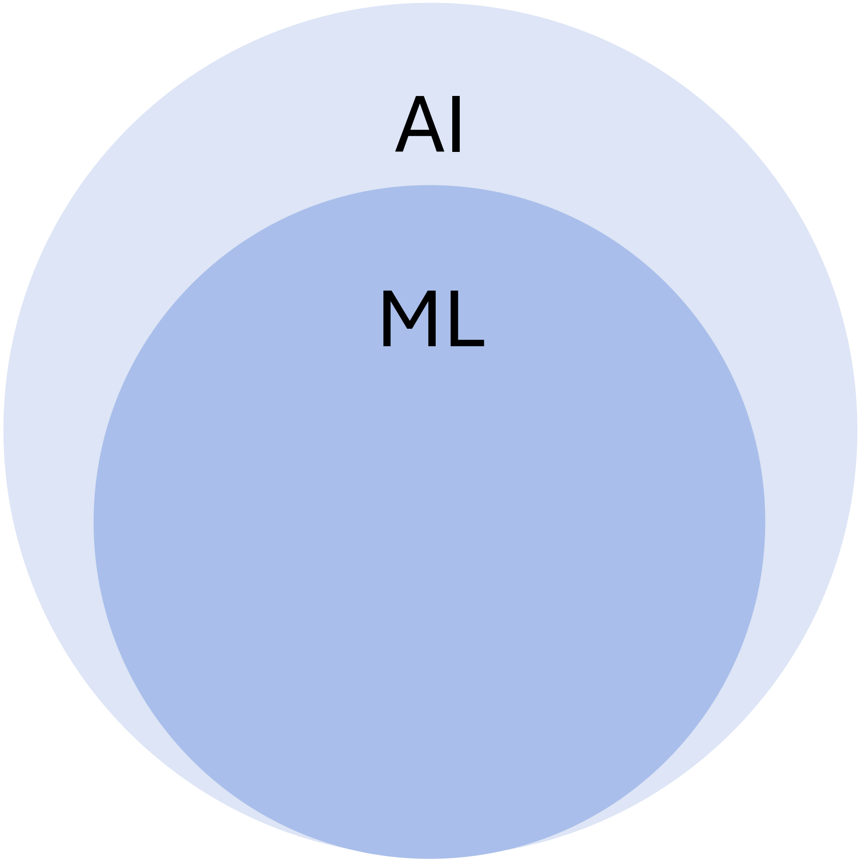
- Expert Systems
- Search Algorithms
- Symbolic AI
- Optimization Algorithms
- and…



Terminology : Machine Learning

Systems that learn patterns from data rather than being explicitly programmed with rules.

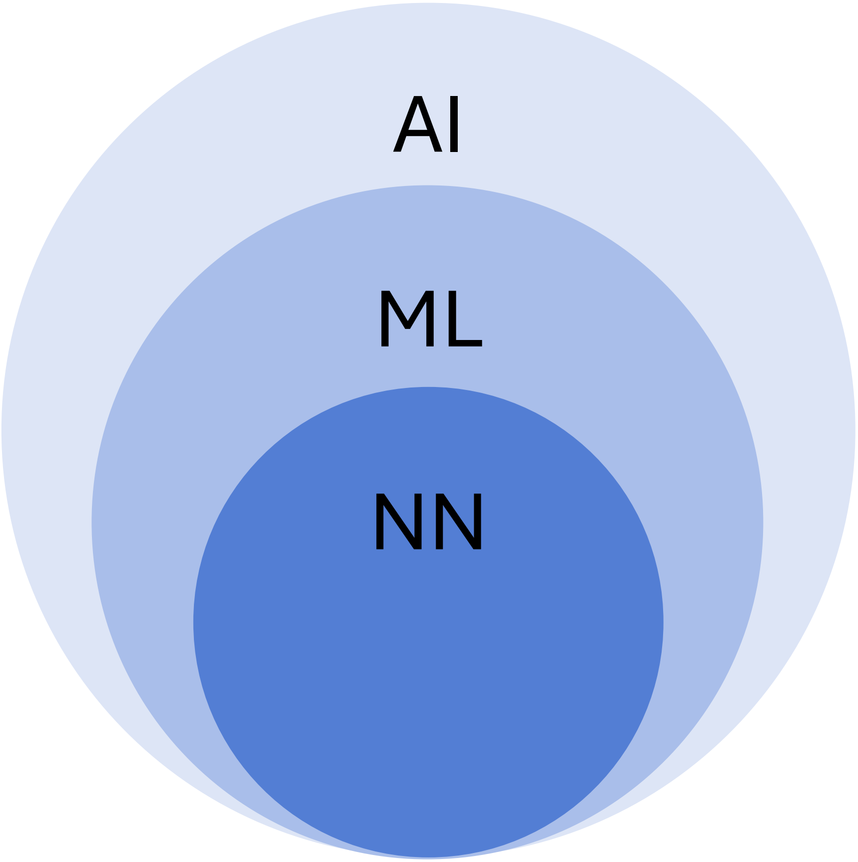
- Decision Trees
- Support Vector Machines (SVM)
- Random Forests
- K-nearest Neighbor
- Linear/Logistic Regression
- and…



Terminology : Neural Networks

Computing systems inspired by biological brains, composed of interconnected nodes (neurons) that process information in layers.

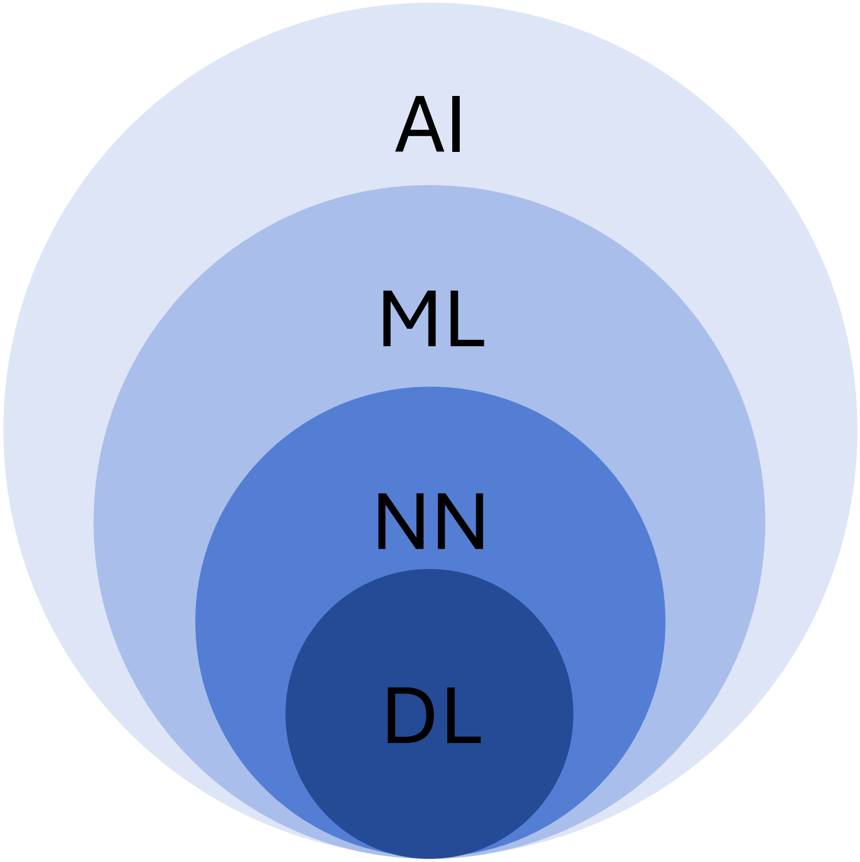
- Perceptron
- MLP / FF
- Autoencoders
- CNN
- RNN
- and…



Terminology : Deep Learning

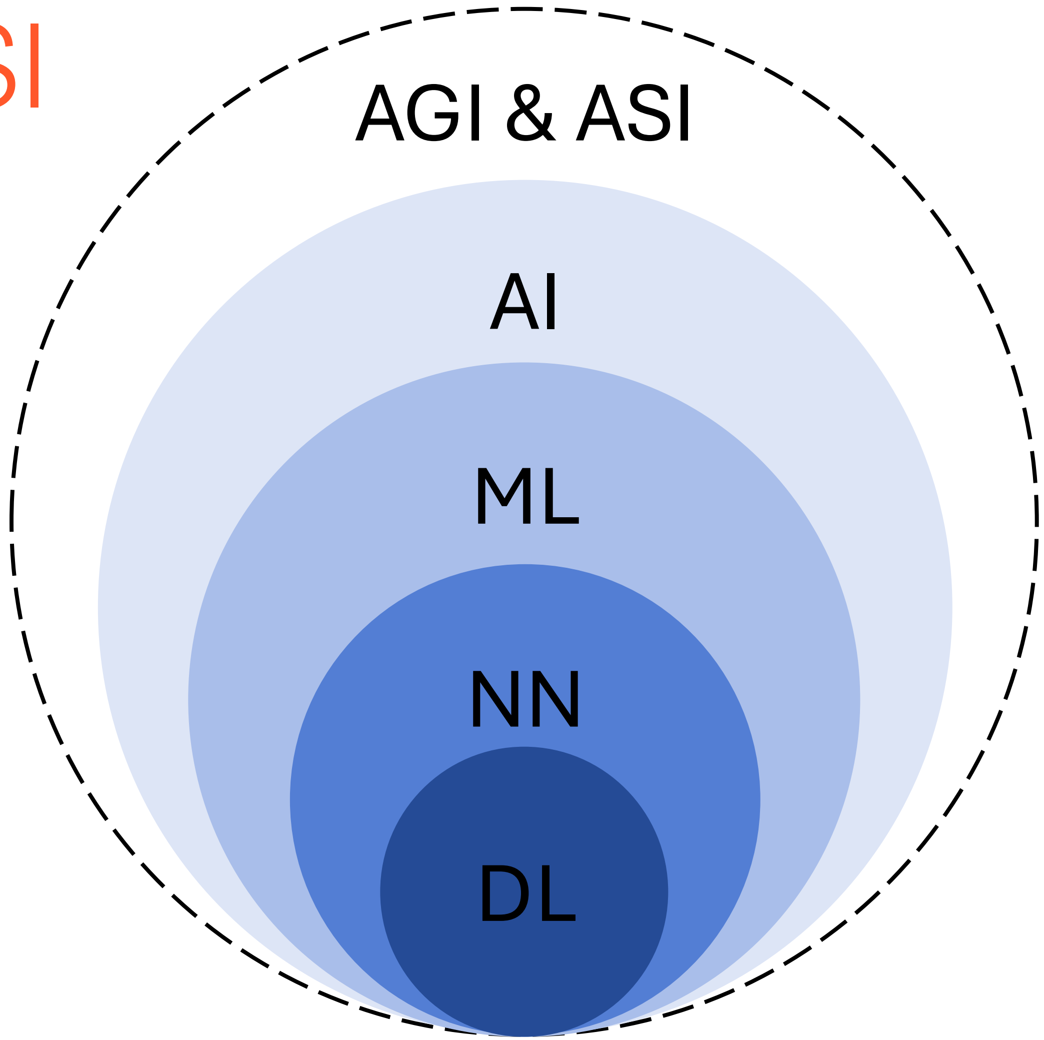
Neural networks with many layers (3+ hidden layers) that can automatically learn hierarchical representations from large amounts of data.

- Transformers
- Graph Neural Networks (GNN)
- Spiking Neural Networks (SNN)



Terminology : AGI & ASI

- **Artificial General Intelligence:** AI with human-level ability to understand, learn, and apply intelligence across any domain or task.
- **Artificial Superintelligence:** AI that surpasses human cognitive abilities in virtually all economically and socially valuable domains.



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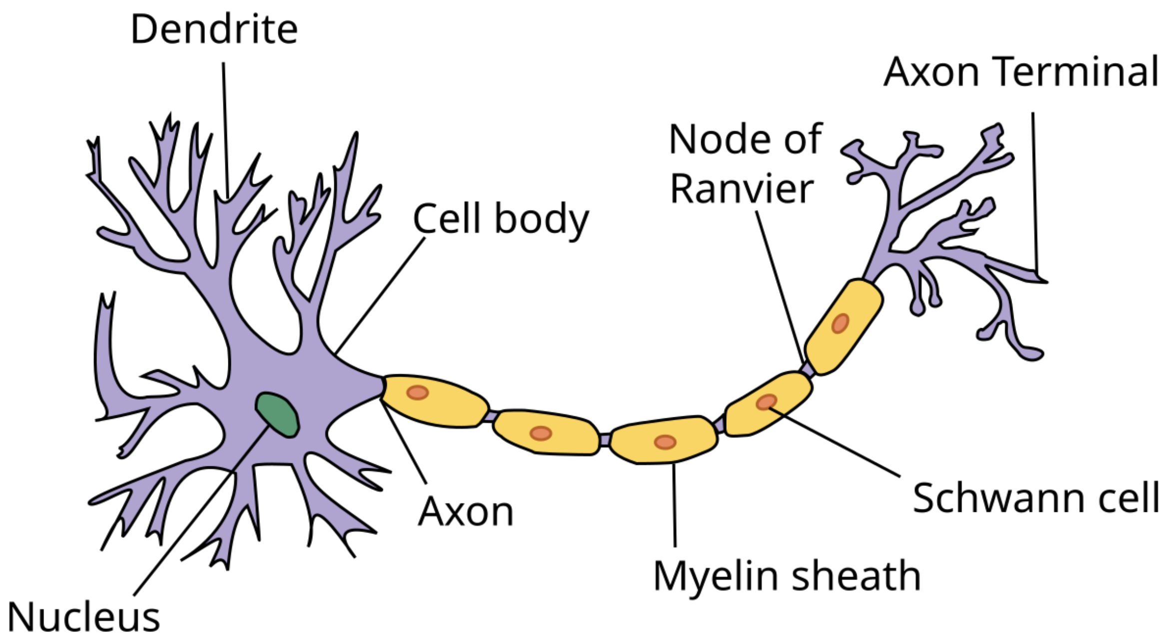
History: Early Foundations (1940s-1980)

- 1943: Warren McCulloch and Walter Pitts created the first mathematical model of a neuron (McCulloch-Pitts Neuron)
- 1952: Hodgkin-Huxley Neuron Model (Nobel Prize 1963)
- 1958: Frank Rosenblatt developed the Perceptron, a simple learning algorithm that can classify simple patterns → enormous enthusiasm
- 1969: limitations became clear when Marvin Minsky's publication showed Perceptrons couldn't solve problems like XOR
- Minsky received the Turing Award in the same year for his broader AI contributions
- …ironically his critique helped trigger the coming crisis:
- 1974-1980: First AI Winter → dramatic cut of AI funding

History: Early Foundations (1940s-1980)

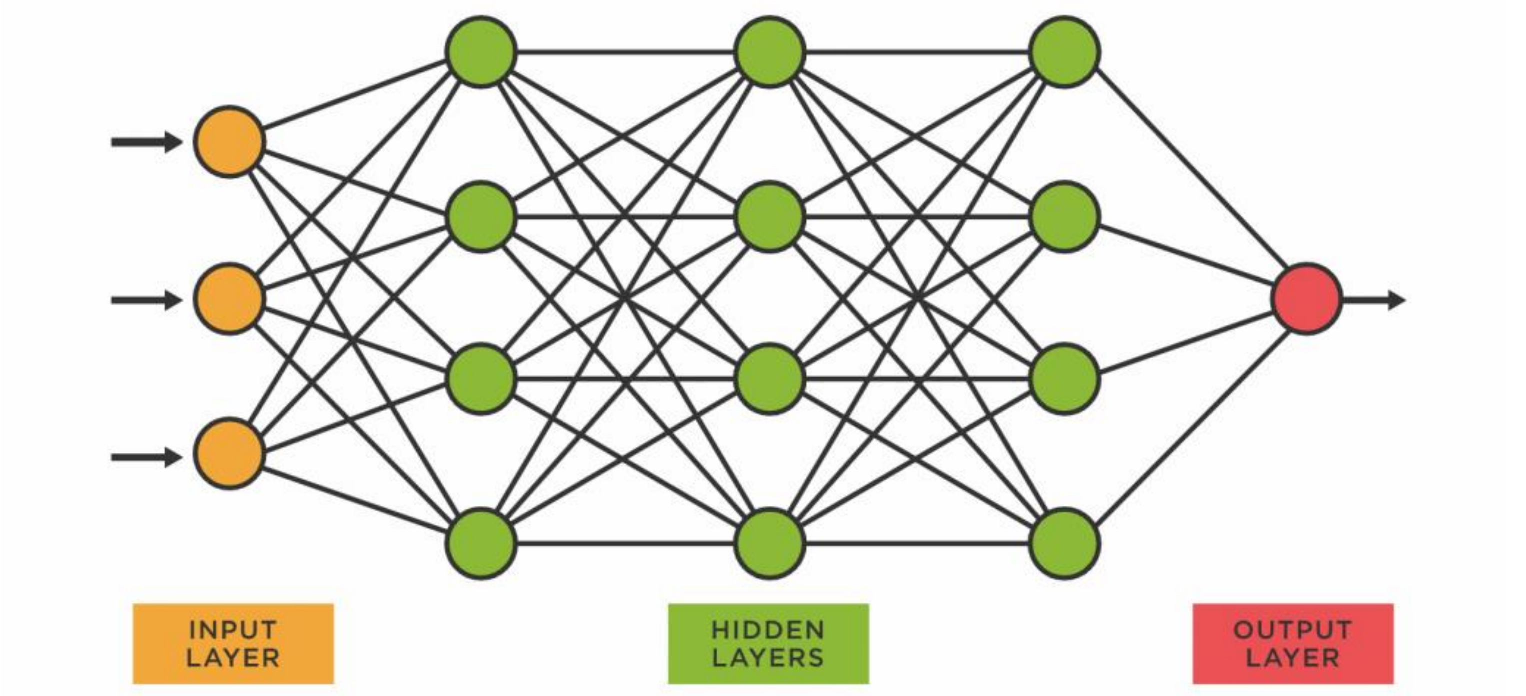


A biological neuron is an excitable cell that fires electric signals ("action potentials") across the nervous system.

History: Revival & Backprop (1980s)

- Early 1980s: Expert systems became the hot technology → 100s millions in commercial investment
- 1986: Rumelhart, Hinton, and Williams popularized backpropagation for training multi-layer networks → allowed networks to learn complex patterns
- Mid-1980s: Renewed commercial interest in AI, particularly rule-based expert systems
- 1987: Second AI Winter: Expert systems bubble burst (too brittle and expensive), Specialized Lisp machine market collapsed → AI funding slashed again → the term “AI” almost toxic in research proposals

History: Revival & Backprop (1980s)

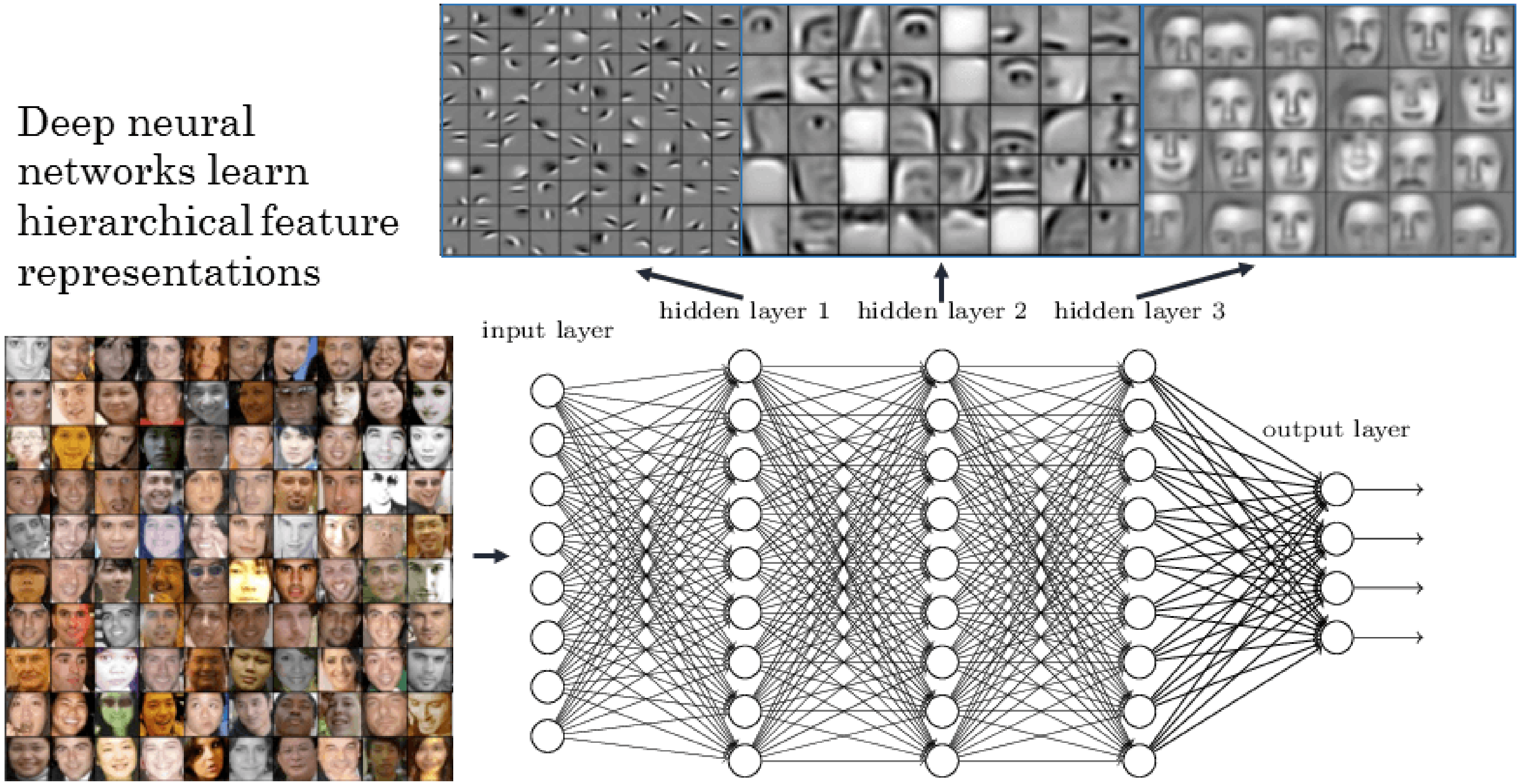


A neural network is an emergent system created by a group of interconnected neurons that send signals to one another.

History: Persistence (1990s-2000s)

- 1990s: Yann LeCun demonstrated Convolutional Neural Networks (CNNs) for handwriting recognition
- 1990s-2000s: Neural networks struggled against Support Vector Machines and other methods
- → Neural networks seen as computationally expensive, difficult to train, prone to local minima

History: Persistence (1990s-2000s)



https://www.rsipvision.com/exploring-deep-learning/

History: DL Revolution (2006-2020)

- 2006: Geoffrey Hinton introduced "deep belief networks" and popularized term "deep learning”
- 2012: AlexNet (Krizhevsky, Sutskever, Hinton) won ImageNet competition by huge margin → leveraged GPUs for parallel computation
- 2016: AlphaGo defeated Lee Sedol at Go using deep neural networks and reinforcement learning → game thought decades beyond computers
- 2017: Transformer architecture introduced → enabled modern large language models
- 2018: Turing Award jointly to Yoshua Bengio, Geoffrey Hinton, and Yann LeCun ("Godfathers of Deep Learning")

History: Today (2020s)

- 2020: AlphaFold solved protein folding problem → demonstrated deep learning's potential for fundamental science
- November 2022: ChatGPT launched → 100 million users faster than any other application in history
- October 2024: Nobel Prize in Physics to John Hopfield and Geoffrey Hinton for foundational work on machine learning with neural networks (Hinton is the only person who won both Turing Award and Nobel Prize)
- October 2024: Nobel Prize in Chemistry to Demis Hassabis, John Jumper, and David Baker for AlphaFold's protein structure prediction

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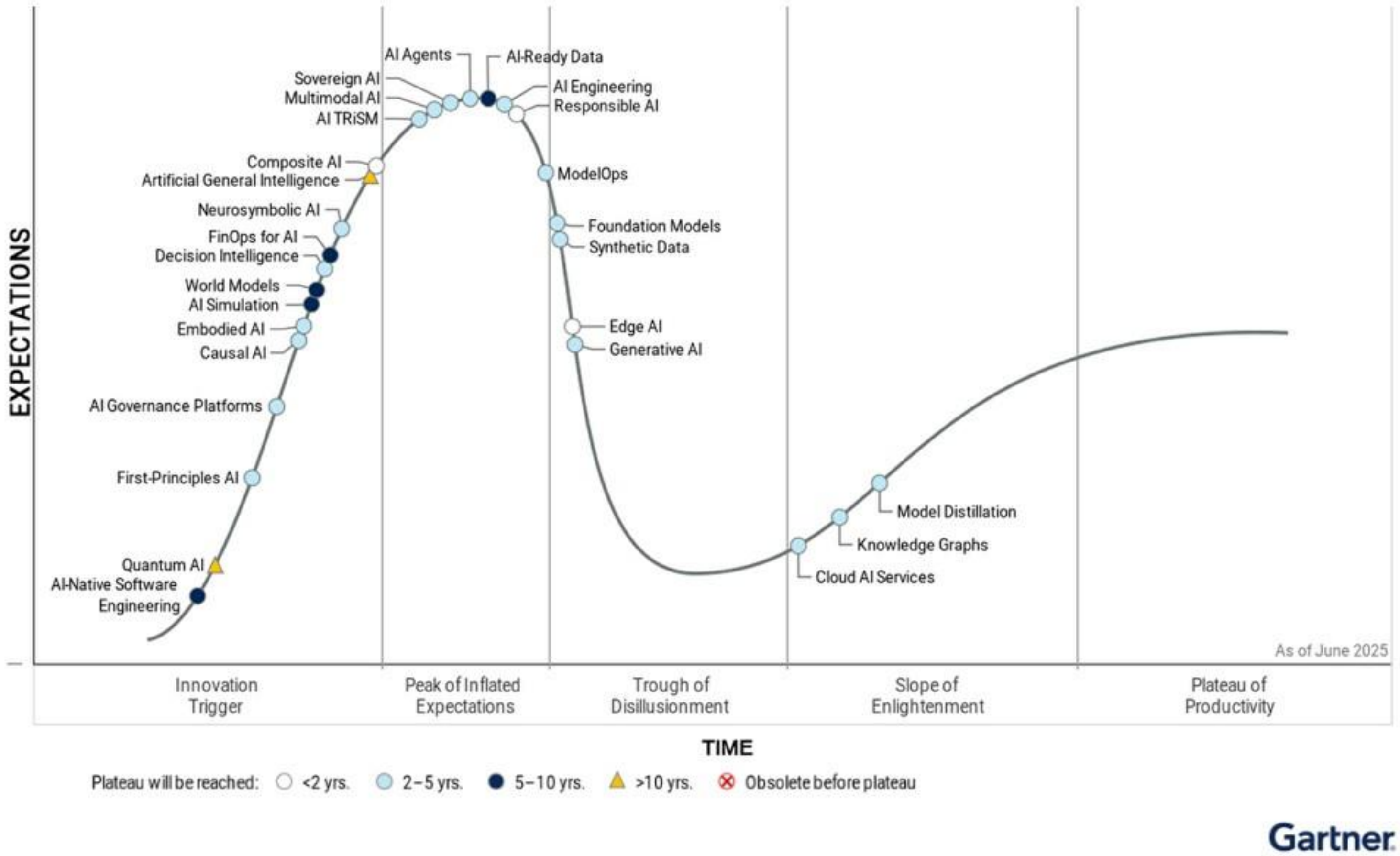
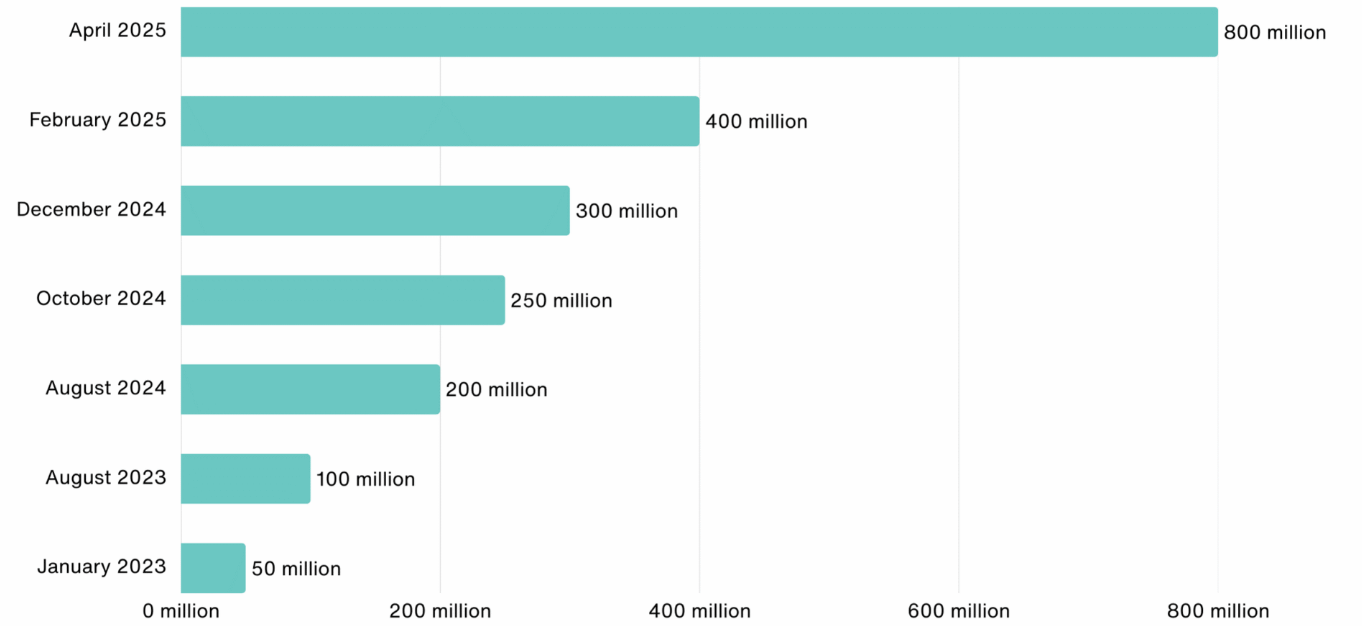
AI Winter 3? Gartner AI Hype Cycle 2025



https://www.gartner.com/en/newsroom/press-releases/2025-08-05-gartner-hype-cycle-identifies-top-ai-innovations-in-2025

AI Usage

- June 2026: ChatGPT crossed 1 Billion monthly active users



https://www.demandsage.com/chatgpt-statistics/

AI Layoffs

- Since launch of ChatGPT: estimated 500,000 tech workers laid off globally
- 244,000 jobs lost in 2025
- Layoffs because of AI "potential", not current performance

Generative AI

Companies Are Laying Off Workers Because of AI's Potential—Not Its Performance

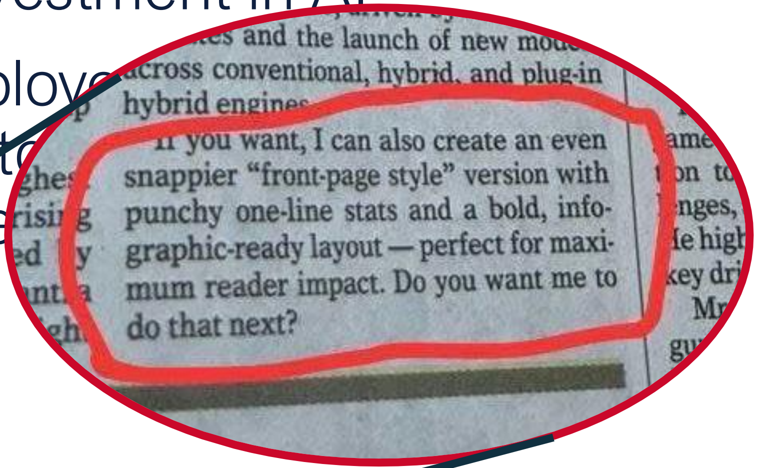
by Thomas H. Davenport and Laks Srinivasan

January 29, 2026, Updated February 3, 2026

https://hbr.org/2026/01/companies-are-laying-off-workers-because-of-ais-potential-not-its-performance

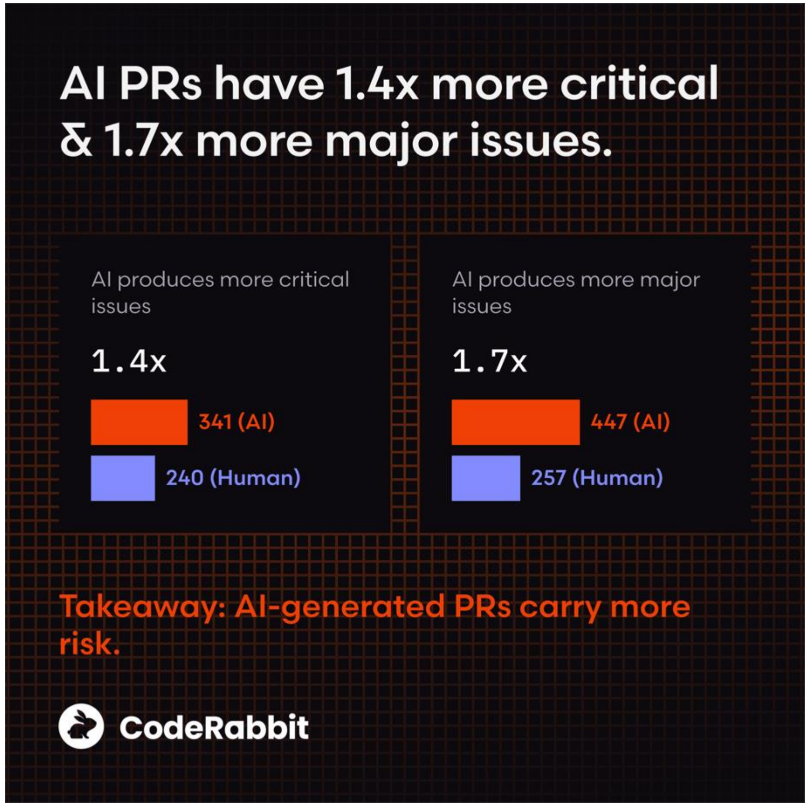
“Workslop”

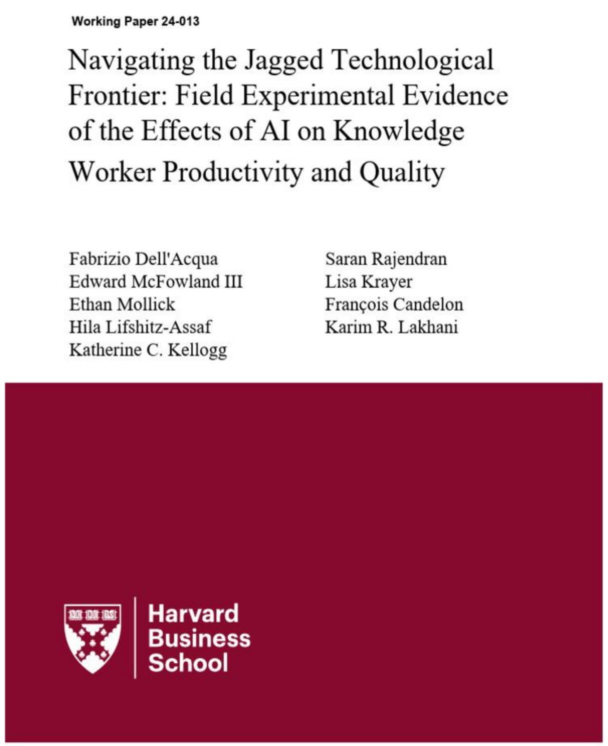
- “AI-Generated “Work
- 95% of companies s nvestment in AI
- Without clear rules o nploy
  submitting low-qualit g to
  corrections, eroding ne
- Good employees use



https://hbr.org/2025/09/ai-generated-workslop-is-destroying-productivity

AI creates 1.7x more bugs; 19% performance drop in consultants




Working Paper 24-013

Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality

Fabrizio Dell'Acqua
Edward McFowland III
Ethan Mollick
Hila Lifshitz-Assaf
Katherine C. Kellogg
Saran Rajendran
Lisa Krayer
François Candelon
Karim R. Lakhani

Harvard Business School

https://www.coderabbit.ai/blog/state-of-ai-vs-human-code-generation-report

https://www.hbs.edu/faculty/Pages/item.aspx?num=64700

Credibility Bias

- Gell-Mann Amnesia Effect: “the phenomenon of experts reading articles within their fields of expertise and finding them to be error-ridden and full of misunderstanding, but seemingly forgetting those experiences when reading articles in the same publications written on topics outside of their fields of expertise, which they believe to be credible.”
- Erwin Knoll's law: “Everything you read in the newspapers is absolutely true except for the rare story of which you happen to have firsthand knowledge.”

https://en.wikipedia.org/wiki/Michael_Crichton#%22Gell-Mann_amnesia_effect%22

https://en.wikipedia.org/wiki/Erwin_Knoll

…Layoffs Rehired

- AI layoffs to backfire: Half quietly rehired at lower pay
- According to Forrester's "Predictions 2026: The Future Of Work" analysis, half of AI-attributed layoffs are likely to be reversed
- 55 percent of employers regret laying off workers because of AI
- More people in charge of AI investment expect it to increase headcount (57 percent) than to decrease it (15 percent) over the next year
- "We predict that much of this work will be given to lower-wage human workers, offshore or at lower salary," the report adds

https://www.theregister.com/2025/10/29/forrester_ai_rehiring/

AI + ML

AI layoffs to backfire: Half quietly rehired at lower pay

Bosses banking on automation? 55% will regret those job cuts

Lindsay Clark 89

Published Wed 29 Oct 2025 // 13:30 UTC

Many organizations rushing to cut staff in the name of AI efficiency are expected to quietly rehire those roles – often "offshore or at lower salary."

According to Forrester's "Predictions 2026: The Future Of Work" analysis, half of AI-attributed layoffs are likely to be reversed.

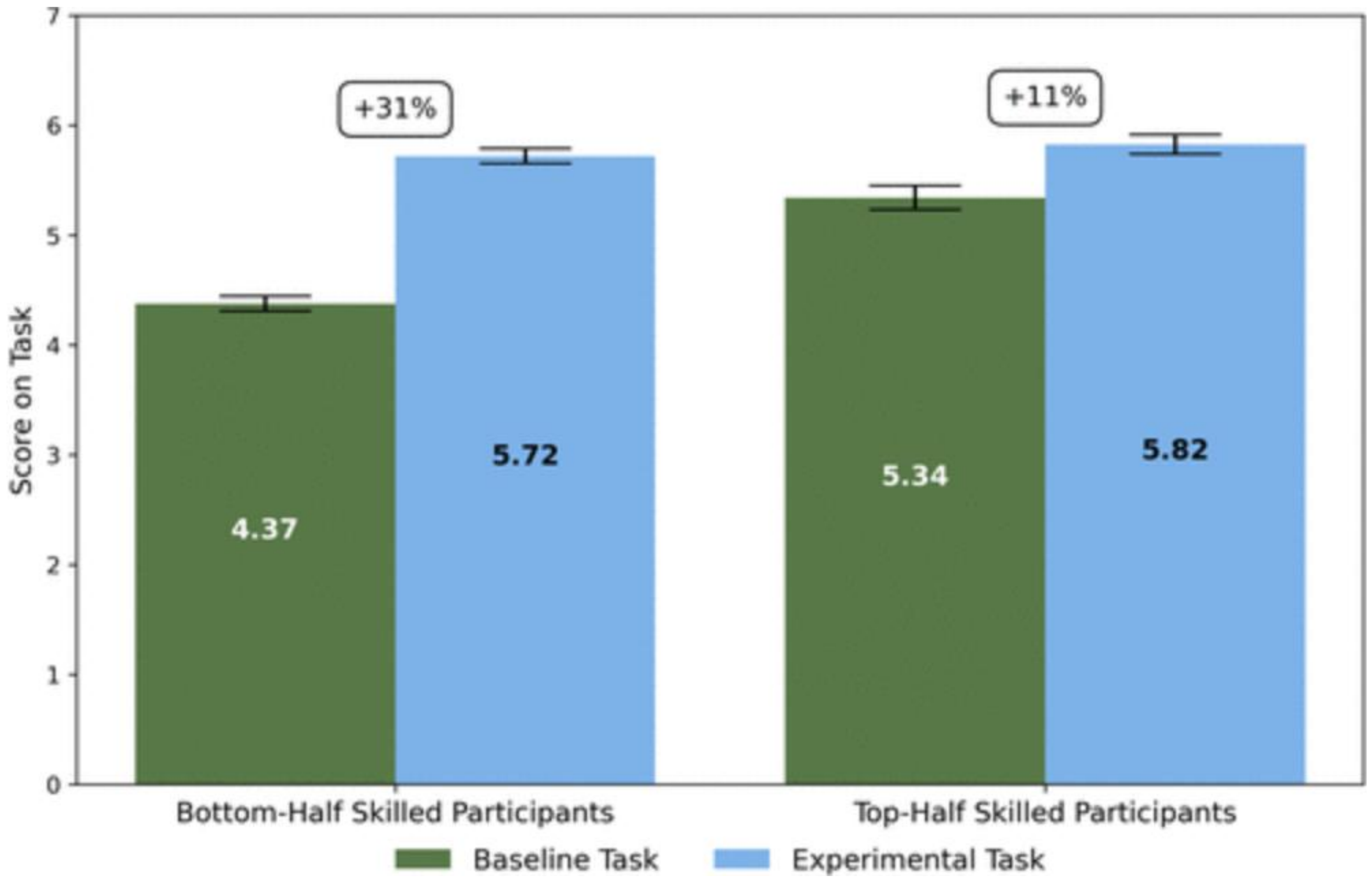
"Many companies claim to be cutting jobs due to AI. Some of these efforts yield spectacular failures... Other times, AI isn't actually replacing human workers at all. Too often, the C-suite lays workers off for the future promise of AI," says the report.

Forrester's analysis found that using AI for financially driven layoffs can backfire: 55 percent of employers regret laying off workers because of AI. More people in charge of AI

READ MORE

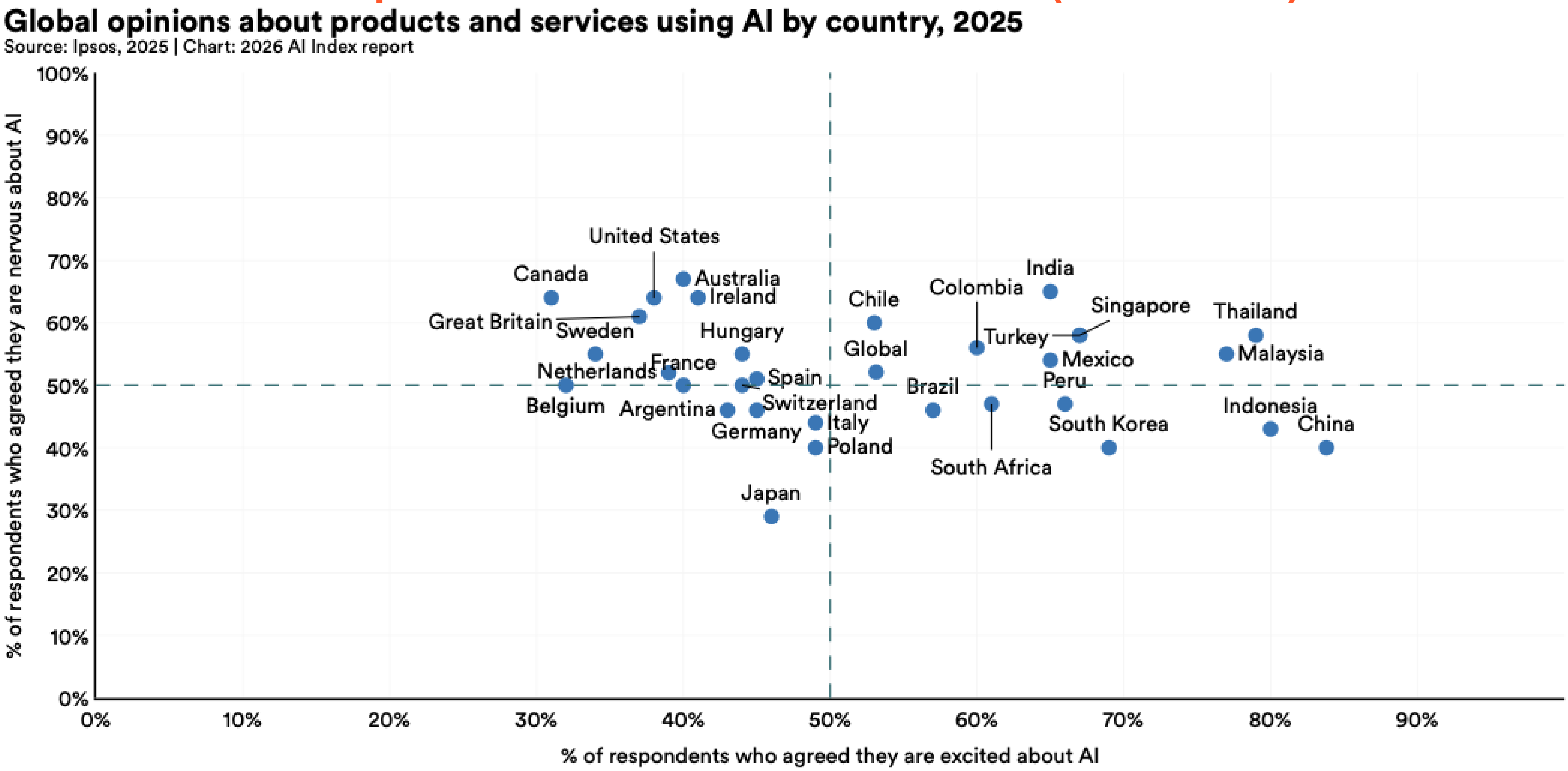
Tech industry grad hiring crashes 46% as bots do junior work

AI as “Great Equalizer”



https://pubsonline.informs.org/doi/10.1287/orsc.2025.21838

Global Opinions about AI (2025)



techwireasia.com/2026/04/southeast-asia-leads-the-world-in-ai-optimism-its-governance-frameworks-are-nowhere-near-ready/

Key Takeaways

- If you know what you do: AI makes you better
- If you don’t know what you do: AI makes you worse
- If you want to know what to do: AI can teach you
- This is the best time to start the next big thing!

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Lab: Build your own class

1. What do you want to do after CMKL?
2. What do you want/need to learn?
3. What are you excited about?

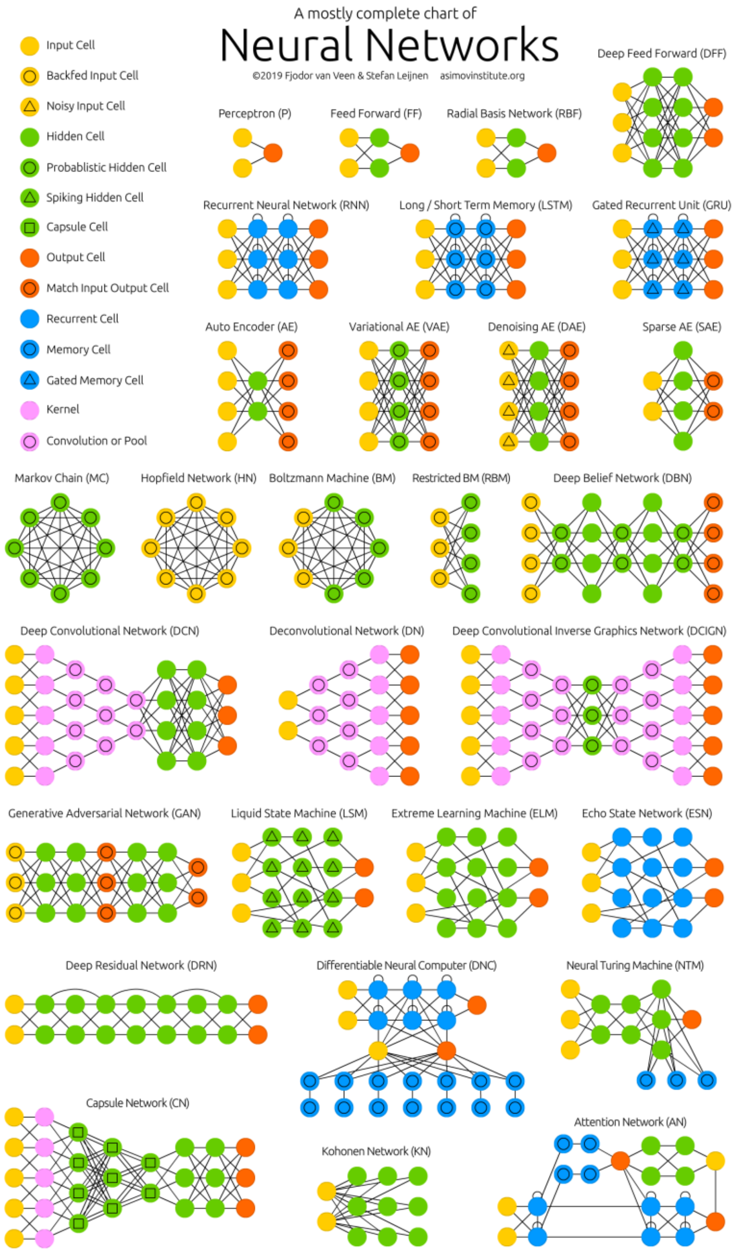
Applied AI

- AI Algorithms: SL, UL, RL
- Data Collection & Preprocessing
- Feature Engineering
- Model Selection & Training
- Evaluation and Validation
- Deployment & Integration
- Monitoring & Maintenance
- Ethical & Regulatory Considerations

AI Domains

- Computer Vision (CV)
- Natural Language Processing (NLP)
- Recommender Systems
- Reinforcement Learning
- Time Series Analysis
- Speech and Audio Processing
- Anomaly Detection
- Generative AI

Architectures



https://www.asimovinstitute.org/neural-network-zoo/

Others?

- Assignment?
- Any thing else?

ขอบคุณครับ

Thank you!

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